

NovaCraft Inc.

Business Intelligence & Data Context

Prepared for the AutoML Platform Demo

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Executive Summary

NovaCraft is a mid-market B2B SaaS company providing cloud-based project management and collaboration tools. Founded in 2018, the company serves approximately 2,500 business customers across 35 countries, with a product suite spanning four subscription tiers (Free, Starter, Professional, Enterprise).

This document describes the business context, data assets, and analytical opportunities available in the NovaCraft mock data package. The data is designed to exercise a full AutoML pipeline: profiling, preprocessing, feature engineering, model training, and evaluation.

Key Business Metrics

- **Total Customers:** ~2,500 active and churned accounts
- **Active Rate:** ~82% of customers remain active
- **Subscription Records:** ~3,200 (1.3 per customer avg.)
- **Support Tickets:** ~6,800 lifetime tickets
- **Usage Tracking:** ~10,700 monthly usage snapshots
- **Marketing Campaigns:** ~500 campaigns (2022–2025)

Business Context

Product & Pricing

NovaCraft offers a tiered SaaS model:

Tier	Target Segment	MRR Range	Mix
Free	Solo / evaluation	\$0	10%
Starter	Small teams	\$29–99	30%
Professional	Mid-size orgs	\$99–499	35%
Enterprise	Large organizations	\$499–2,500+	25%

Revenue is primarily driven by seat-based pricing with annual and monthly billing cycles. Enterprise accounts negotiate custom pricing with dedicated account managers.

Customer Base

Customers span 12 industries, with Technology, Healthcare, and Finance representing the largest segments. Company sizes range from startups (1–10 employees) to large enterprises (1,000+), with the majority in the 11–200 employee band.

Acquisition channels include organic search (25%), paid search, social media, referrals, direct signups, partner channels, and content marketing.

Stakeholders

- **Product Team** — Feature adoption, usage patterns, NPS analysis
- **Customer Success** — Churn prediction, health scoring, escalation risk
- **Sales** — MRR forecasting, upsell/cross-sell identification
- **Marketing** — Campaign ROI, channel attribution, lead quality
- **Support** — Resolution time optimization, agent performance, CSAT drivers

Data Dictionary

customers.csv

Core customer master data with 14 columns and ~2,530 rows (including 30 intentional duplicate rows for deduplication testing).

Column	Type	Missing	Description
customer_id	string	0%	Primary key (NC-XXXXX)
company_name	string	0%	Business name (high cardinality)
industry	string	0%	Business vertical (12 categories)
company_size	string	0%	Ordinal employee band
country	string	0%	Country of headquarters
signup_date	date	0%	Account creation date
plan_tier	string	0%	Current subscription tier
annual_revenue_usd	float	8%	Annual revenue (right-skewed, outliers)
employee_count	integer	5%	Employee headcount (right-skewed)
acquisition_channel	string	0%	Marketing attribution channel
account_manager	string	12%	Assigned account manager
is_active	boolean	0%	Classification target
region_code	string	0%	Constant column (all "GLOBAL")
data_source	string	0%	Ingestion source (<20 unique)

subscriptions.csv

Subscription lifecycle records with 12 columns and ~3,200 rows.

Column	Type	Missing	Description
subscription_id	string	0%	Primary key
customer_id	string	0%	FK to customers
plan_name	string	0%	Plan tier at subscription time
billing_cycle	string	0%	monthly or annual
mrr_usd	float	3%	Monthly recurring revenue (right-skewed)
start_date	date	0%	Subscription start
end_date	date	39%	Null if still active
cancellation_reason	string	51%	Reason for churn

Column	Type	Missing	Description
discount_pct	float	15%	Applied discount (outliers present)
seats_purchased	integer	0%	License seat count
payment_method	string	0%	Payment type (4 categories)
auto_renew	boolean	2%	Auto-renewal flag

support_tickets.csv

Customer support interactions with 11 columns and ~6,800 rows.

Column	Type	Missing	Description
ticket_id	string	0%	Primary key
customer_id	string	0%	FK to customers
created_at	date	0%	Ticket creation timestamp
resolved_at	date	20%	Resolution date (correlated missing)
category	string	0%	8 ticket categories
priority	string	0%	low / medium / high / critical
resolution_hours	float	20%	Hours to resolve (correlated missing)
satisfaction_score	integer	25%	CSAT 1–5 (left-skewed)
agent_id	string	0%	Support agent identifier
escalated	boolean	0%	Classification target
description_length	integer	0%	Ticket text length (right-skewed)

usage_metrics.csv

Monthly product usage aggregates with 13 columns and ~10,700 rows.

Column	Type	Missing	Description
customer_id	string	0%	FK to customers
month	date	0%	First-of-month date
active_users	integer	0%	Monthly active users
total_logins	integer	0%	Total login count (right-skewed)
projects_created	integer	0%	New projects created
tasks_completed	integer	0%	Completed tasks (outliers)
storage_used_gb	float	7%	Storage consumption in GB
api_calls	integer	0%	API requests (extreme right-skew)
integrations_active	integer	0%	Active third-party integrations
avg_session_minutes	float	10%	Average session duration
feature_adoption_pct	float	5%	Percentage of features used
nps_response	integer	50%	NPS score 0–10 (critical missing)

Column	Type	Missing	Description
export_count	integer	0%	Data exports (near-constant, 95% = 0)

marketing_campaigns.csv

Campaign performance data with 10 columns and ~505 rows (including 5 duplicates).

Column	Type	Missing	Description
campaign_id	string	0%	Primary key
campaign_name	string	0%	Descriptive name (high cardinality)
channel	string	0%	Marketing channel (6 categories)
start_date	date	0%	Campaign start date
end_date	date	0%	Campaign end date
budget_usd	float	0%	Campaign budget (right-skewed)
leads_generated	integer	0%	Leads produced (outliers)
conversions	integer	0%	Regression target
cost_per_lead	float	8%	Cost per lead metric
target_industry	string	0%	Target vertical (12 + “All”)

Data Quality Assessment

The datasets contain intentionally injected data quality issues to exercise the AutoML platform’s profiling and preprocessing capabilities:

Issue	Affected Columns	Severity
Missing values (~5%)	employee_count, feature_adoption_pct	Low
Missing values (~10–25%)	account_manager, discount_pct, satisfaction_score	Medium
Missing values (~40–60%)	end_date, cancellation_reason, nps_response	High
Outliers (IQR)	annual_revenue, resolution_hours, api_calls	Medium
Right-skewed distributions	mrr_usd, api_calls, tasks_completed, budget_usd	Medium
Left-skewed distribution	satisfaction_score	Low
Constant column	region_code (all “GLOBAL”)	High
Near-constant column	export_count (95% zeros)	Medium
Duplicate rows	customers (30), marketing_campaigns (5)	Medium
Type conversion candidate	data_source (<20 unique strings)	Low
Correlated missingness	resolved_at ↔ resolution_hours	Medium

Recommended Analyses

Customer Churn Prediction

Target: `is_active` (binary classification)

Join customers with aggregated usage metrics and support ticket counts. Key features: tenure (months since signup), plan tier, support ticket frequency, average satisfaction score, usage trends, and subscription changes.

Revenue Forecasting

Target: `mrr_usd` (regression)

Predict monthly recurring revenue from customer profile, company size, industry, usage intensity, and feature adoption metrics.

Support Escalation Prediction

Target: `escalated` (binary classification)

Use ticket metadata (category, priority, description length) combined with customer history to predict which tickets will require escalation.

Marketing Campaign Optimization

Target: `conversions` (regression)

Model campaign conversion rates as a function of channel, budget allocation, target industry, campaign duration, and historical performance.

Cross-Table Feature Engineering

The relational structure enables rich derived features:

- Tickets per customer per month
- Average satisfaction trend over time
- Usage growth rate (month-over-month)
- Subscription upgrade/downgrade patterns
- Days since last support interaction

This document accompanies the NovaCraft mock data package for the AutoML platform. Data is synthetically generated for demonstration purposes and does not represent any real company or individuals.